

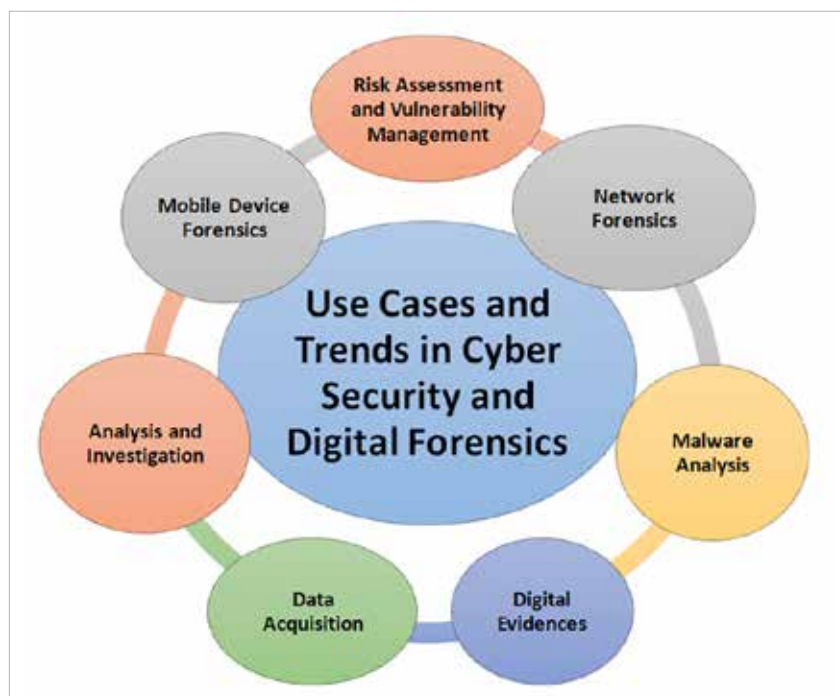


Figure 2: Use cases and trends in cyber security and digital forensics

means for security professionals to exchange credible threat data across a trusted cadre, free from the risk of tampering. Security protocols are further automated through smart contracts, which must be adhered to, meaning that few humans are involved and human error is eliminated. In general, blockchain strengthens the resilience to cyber threats.

Using a blockchain for cybersecurity applications involves:

- Data integrity
- Identity and access management
- Data protection
- Secure communication and transactions
- Ransomware protection

Using a blockchain for digital forensics applications entails:

- Chain of custody (CoC) management
- Timestamping and evidence authentication
- Smart contracts for compliance
- Decentralised threat intelligence

Blockchain technology addresses data integrity and protection

by using smart contracts. The immutability of the blockchain secures sensitive information like medical records, financial transactions and legal documents from nefarious tampering or erasure. For instance, the government in Estonia uses the blockchain to secure all its sensitive data.

Blockchain is also being used to protect smart grids and other critical infrastructure. Blockchain based networks allow organisations to share real-time threat information in a secure environment (without the risk of data tampering or data leakage) while ensuring that all incidents, events and responses are written to a permanent shared ledger to maintain accurate records, and to build a feedback loop based on the exchange of threat intelligence.

Frameworks for collaboration to facilitate sharing of cyber threat data in real-time using blockchain technology to counter global cyber threats are being established by many government and corporate agencies. As cyber-attacks become more advanced, the blockchain is

emerging as a key technology, offering a game-changing platform for businesses, governments and individuals to secure their digital assets in a transparent, trustless and tamper-proof way.

Using blockchain for digital evidence management

The role of digital evidence in modern law enforcement, cybersecurity investigations and court cases cannot be overstated. Nonetheless, it poses a significant challenge with respect to integrity, authenticity and chain of custody. Conventional evidence management stores data in centralised databases, prone to tampering, unauthorised access and data corruption. Blockchain technology is a decentralised, immutable and transparent ledger that has the potential of transforming how digital evidence is managed by securely indenting every stage of the digital evidence life cycle.

In practice, a national law enforcement agency struggles to ensure the integrity of digital evidence including emails, surveillance footage, and forensic reports. There is no secure tracking system, which raises concerns of unauthorised changes and losing that critical data. To combat these challenges, the agency can adopt a ledger-based evidence management platform that logs each evidence with a unique cryptographic hash key. This hash is kept in a distributed ledger so that if any evidence is modified, it will no longer match the hash value, indicating that the evidence has been tampered with.

Transparency and auditability of the chain of custody: A major advantage of this system is an easily auditable chain of custody. Any access, change, or transfer of evidence is automatically recorded as a new transaction on the blockchain, creating an immutable activity log. It enables legal professionals, law enforcement officers and forensics analysts to authenticate the provenance of evidence directly, without